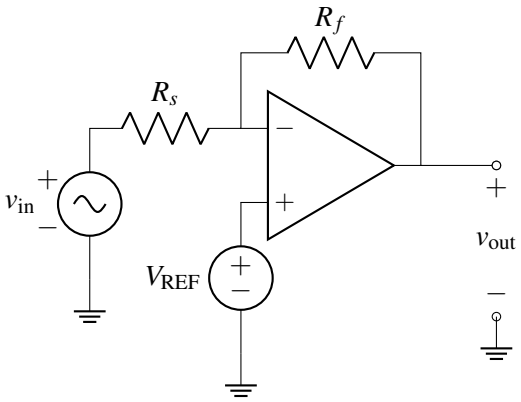
CSM
Fall 2026

EE 64
Worksheet 9

1. Negative Feedback

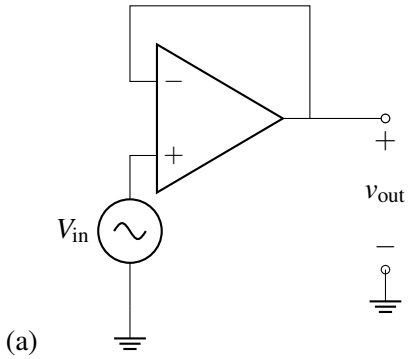
Are the following circuits in negative feedback? Explain your reasoning.



(a)

(b) Suppose that we switched the negative and positive signs on the op-amp. Is this circuit still in negative feedback?

Let's observe one of the most important op-amp configurations: the unity gain buffer!



(a)

(b) Knowing the properties of an op-amp, and that the unity gain buffer has a gain of 1, what are some possible uses of unity gain buffers?

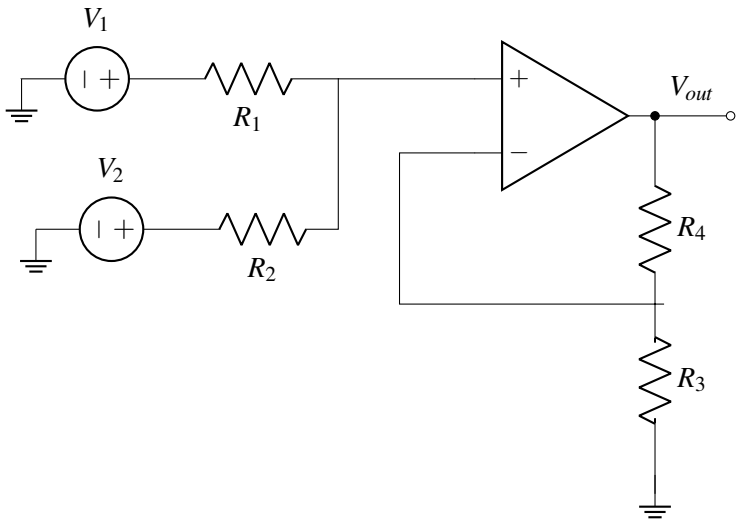
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2. Voltage Summers

Learning Goal: This problem uses basic circuit analysis techniques to find the response of a summer circuit.

Relevant Notes: *Note 19* goes different op-amp circuit topology and corresponding derivations.

(a) Calculate V_{out} in terms of V_1 and V_2. Assume that $R_1 = R_2$. Use superposition.



(b) What values should we select for R_1, R_2, R_3, and R_4 such that $V_{out} = V_1 + 2V_2$?

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3. Multi-stage Amplifier

Learning Goal: The objective of this problem is to understand how multiple stages of op-amp circuits can be used to achieve a specific circuit gain.

Relevant Notes: *Note 19 Section 19.5* goes over inverting and non-inverting amplifiers.

(a) What is the range of values that we can scale V_{in} by when using a non-inverting op amp? (What are possible values for the gain?)

(b) What is the range of values that we can scale V_{in} by when using an inverting op amp? (What are the possible values for the gain?)

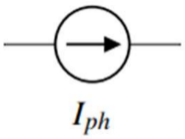
(c) Can you design a single inverting/ non-inverting amplifier with circuit gain $G = 0.5$? If not, what range of gain values is not reachable using a single inverting op amp or a single non-inverting op amp?

(d) How would you construct a circuit using inverting/ non-inverting amplifiers so that the overall circuit gain is $G = 0.5$?

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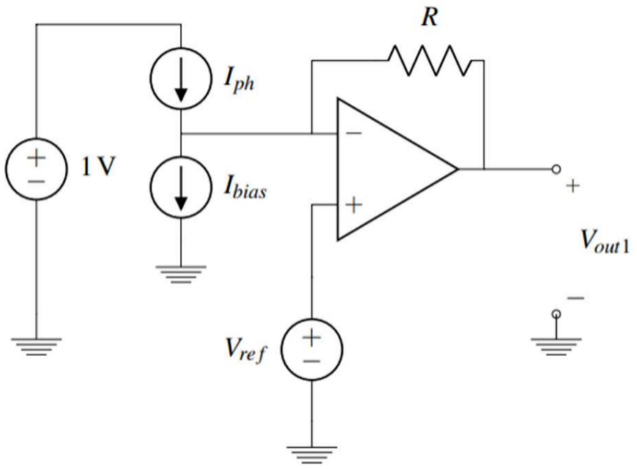
4. Op-Amp Analysis

After finishing EE64, you get the opportunity to work with Professor Vladimir. His group designs computer chips that communicate with light, and your job is to design the receiver. Your receiver connects to a photodetector, that can be modeled as current source.



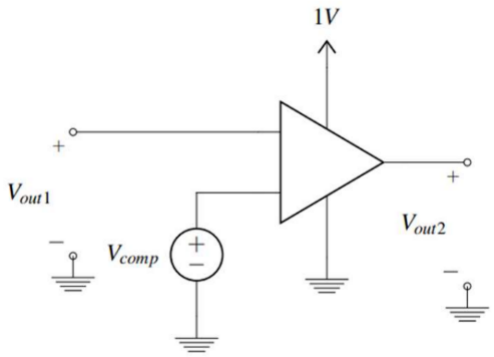
When light is shining on the receiver, the current becomes $I_{ph} = 1\mu A$. Otherwise, with no light shining on the receiver the current is $I_{ph} = 0A$. The circuit should output $0V$ to the processor when there is light shining on the receiver, and output $1V$ to the processor when there is no light shining on the receiver.

(a) Your friend suggests the following circuit to turn the current into a voltage. Find the output voltage, V_{out} as a function of I_{ph}, and find a value of R such that the circuit outputs $0V$ when $I_{ph} = 1\mu A$.
Here, $I_{bias} = 0.5\mu A$ and $V_{ref} = 0.5V$. Show your work.



(b) After building the circuit and testing it, you find that I_{ph} varies as you get closer or further from light. This causes the output voltage of your circuit to not be precisely $0V$ or $1V$.
However, the processor needs exactly $0V$ when light is shining, and $1V$ when no light is shining. Vlad advises you add a comparator to your design. Let $I_{ph,light}$ denote the minimum current indicating there is enough light.
Finish the circuit below by calculating the value of V_{comp} for the comparator. Also, select the polarity of the comparator by drawing a '+' or '-' in corresponding boxes in the diagram below.

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